A centralized, intelligent healthcare platform designed to reduce fragmentation in patient medical information and improve communication between patients and healthcare professionals.

The platform enables patients to maintain digital medical records and uses AI to organize and summarize medical information for easier review by doctors and patients. It also connects records with appointment booking, online/offline consultation and telemedicine.

AI-POWERED ELECTRONIC MEDICAL RECORD & DIGITAL HEALTHCARE PLATFORM

EMR PROJECT — MASTER DOCUMENT

Short Name: Smart EMR

Core Idea

1. Problem Statement

Original: Doctors and patients struggle with fragmented medical data, long waiting times, and difficulty accessing quality medical services.

Expanded: Healthcare information is often distributed across hospitals, clinics, diagnostic laboratories, paper prescriptions, PDFs, images and separate applications. This makes it difficult for patients to maintain a complete history and for doctors to quickly understand previous treatment.

- Repeated tests
- Repeated documentation
- Longer consultation times
- Delayed diagnosis
- Difficulty understanding previous treatment
- Medication-history gaps
- Poor continuity of care
- Increased administrative workload
- Difficulty accessing specialists
- Long hospital waiting times

Core question: How can a patient's healthcare information and healthcare journey be made accessible, organized, understandable and actionable while keeping the patient in control of their data?

2. Background

Healthcare is increasingly digital, but digitization alone does not eliminate fragmentation. India's Ayushman Bharat Digital Mission (ABDM) promotes a connected digital-health ecosystem involving components such as ABHA, the Healthcare Professional Registry, the Health Facility Registry and interoperable digital-health infrastructure.

The project therefore aims to provide an intelligent EMR layer that organizes, summarizes and makes healthcare information useful while keeping interoperability and consent in mind.

3. Vision

To create a connected, intelligent and patient-centric healthcare ecosystem where medical information is available when needed, understandable to doctors and patients, and securely shared with consent.

4. Mission

- Digitize fragmented medical information.
- Create a longitudinal patient medical history.
- Reduce the time doctors spend searching through old records.
- Help patients understand their medical documents.
- Improve continuity of healthcare.
- Reduce unnecessary repetition of medical information and investigations.
- Improve access to healthcare through digital consultations and appointment management.
- Build a privacy-first healthcare platform.
- Create a foundation that can eventually integrate with India's digital-health ecosystem.

5. Objectives

- **Centralized Medical Records:** Provide patients with a secure digital location for their medical information.
- **Intelligent Medical History:** Automatically organize records chronologically.
- **AI-Powered Report Summarization:** Convert lengthy medical reports into concise summaries.
- **Faster Clinical Understanding:** Help doctors quickly review relevant previous information.
- **Appointment Management:** Allow patients to find doctors and book consultation slots.
- **Telemedicine:** Enable remote consultations.
- **Patient Understanding:** Provide patient-friendly explanations without replacing a doctor.
- **Secure Data Management:** Use authentication, authorization, encryption and consent mechanisms.

6. Target Users

Patients

- Create an account
- Maintain medical profile
- Upload reports
- View previous reports and prescriptions
- Track medical history
- Search and book doctors
- Request online consultation
- Share selected records
- View AI summaries
- Manage consent

Doctors

- Manage professional profile
- View appointments

- Access authorized patient records
- Review medical history
- Read AI summaries
- Add consultation notes
- Generate prescriptions
- Update treatment information
- Conduct online consultations

Hospitals / Clinics — Future

- Manage doctors and appointments
- Maintain patient records
- Upload diagnostic results
- Integrate existing systems
- Exchange authorized information

7. Proposed Solution

Patient Health Record

Stores patient profile, reports, prescriptions, diagnoses, allergies, medications and consultations.

AI Medical Intelligence

Processes medical documents and generates summaries, findings, timelines and discussion points.

Healthcare Services

Provides doctor discovery, appointment booking, telemedicine and offline consultation scheduling.

Security & Interoperability

Provides authentication, role-based access, consent, encryption, audit logs and future ABDM/FHIR integration.

8. Core Features

- Patient registration and secure authentication.
- Digital Electronic Medical Record.
- Medical history and longitudinal timeline.
- Upload and management of PDF/image reports.
- AI-powered report analysis and summarization.
- Patient-friendly report explanation.
- Doctor dashboard and patient overview.
- Appointment and slot booking.
- Online and offline consultations.
- Digital prescriptions.
- Consent-based record sharing.
- Audit logging and security controls.

9. Electronic Medical Record

The EMR is the central component of the platform.

Personal Information

Name, age, gender, contact details

Medical History

Previous illnesses, surgeries, chronic conditions, hospitalizations

Medication History

Medicine, dosage, duration, prescribing doctor, status

Allergy Information

Drug, food and other known allergies

Diagnostic Reports

Blood tests, urine tests, imaging, pathology and other laboratory reports

Consultation History

Date, doctor, symptoms, diagnosis, treatment, prescription and follow-up

10. Patient Medical Timeline

Instead of showing records as disconnected files, the system creates a chronological healthcare journey. Example: January — consultation; February — blood test; March — specialist consultation; April — prescription; May — follow-up.

11. AI Medical Report Analysis

```
Upload Report
     ↓
Document Processing
     ↓
Text Extraction / OCR
     ↓
Medical Information Identification
     ↓
AI Analysis
     ↓
Structured Summary
     ↓
Patient View + Doctor View
```

The AI can identify and organize patient information, test information, important findings, reported abnormal values and a concise summary. The analysis can also update the medical timeline.

Patient-Friendly Explanation

The platform can provide a simplified explanation of what a report contains. It should not claim a definitive diagnosis. Appropriate wording is: “This result may require discussion with a qualified healthcare professional.”

12. Doctor Dashboard

- Today's appointments
- Patient search

- Patient overview
- Allergies and current medications
- Previous consultations
- Recent reports
- AI summary
- Original reports
- Consultation notes
- Prescription generation

13. Patient Dashboard

- Health overview
- Medical timeline
- Reports
- Appointments
- Consultations
- Prescriptions
- AI insights
- Consent management

14. Appointment Booking

Workflow: Select doctor → select date → select slot → confirm appointment. The appointment becomes part of the patient's healthcare journey.

15. Online Consultation

```
Book Online Appointment
          ↓
Doctor Accepts
          ↓
Consultation Room
          ↓
Doctor Reviews Authorized EMR
          ↓
Consultation
          ↓
Prescription / Notes
          ↓
Record Stored in Patient History
```

16. Offline Consultation

Patient finds a doctor, selects a hospital/clinic and slot, and books a physical consultation. The appointment is stored in the same patient history.

17. Digital Prescription

After consultation, the doctor can generate medicine name, dosage, frequency, duration, instructions and follow-up date. The prescription is stored in the patient's record.

18. Record Sharing & Consent

Patients should be able to share selected records rather than their entire history. Example: share a blood report and previous prescription with a particular doctor.

```
Patient
  ↓
Select Doctor
  ↓
Select Records
  ↓
Grant Access
  ↓
Consultation
  ↓
Revoke / Expire Access
```

19. Security Architecture

Authentication

Secure login, password hashing, session/token authentication and optional OTP/MFA.

Authorization

Role-based access: patients access their own records; doctors access records only when appropriately authorized; administrators manage platform operations without automatic unrestricted medical-record access.

Encryption

Use HTTPS/TLS for data in transit and appropriate encryption for sensitive data at rest. Passwords must never be stored in plain text.

Audit Logs

Record important access events such as user, action, resource and timestamp.

20. Privacy

The platform should follow a Privacy-by-Design philosophy. For the prototype, demonstrate consent, data minimization, access control, secure authentication, encryption, audit logging and appropriate retention/deletion concepts. Production deployment requires professional legal and regulatory review.

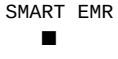
21. Interoperability

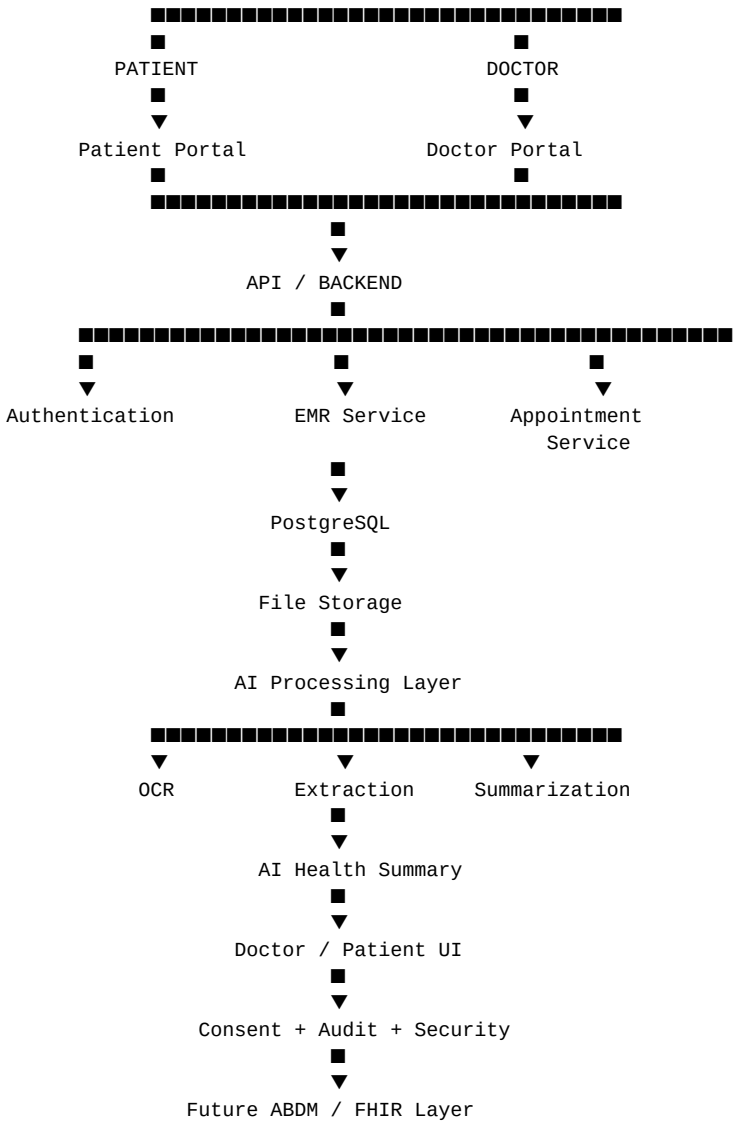
The long-term architecture should support interoperability through standards such as HL7 FHIR and relevant ABDM ecosystem interfaces. The hackathon prototype should present these as an interoperability direction rather than claiming full production compliance.

22. Potential ABDM Integration

- ABHA
- Healthcare Professional Registry
- Health Facility Registry
- Health Information Exchange and Consent Manager
- FHIR-based data exchange

23. High-Level System Architecture





24. Suggested Technology Stack

- **Frontend:** React + Tailwind CSS (or HTML/CSS/JavaScript for a lightweight prototype)
- **Backend:** Django + Django REST Framework
- **Database:** PostgreSQL; SQLite for rapid prototyping
- **Storage:** Object/file storage for medical documents
- **AI:** OCR + NLP/LLM pipeline
- **Communication:** Secure REST APIs

25. AI Pipeline

```
Medical PDF/Image
       ↓
OCR / Text Extraction
       ↓
Text Cleaning
       ↓
Medical Information Extraction
       ↓
LLM / NLP Model
       ↓
Structured JSON
       ↓
```

```
Summary
       ↓
Doctor / Patient Dashboard
```

26. Example AI Output

```
{
  "report_type": "Blood Test",
  "date": "2026-08-20",
  "key_findings": [],
  "abnormal_values": [],
  "summary": "",
  "doctor_discussion_points": []
}
```

27. Database Design

User

id, name, email, password_hash, role, created_at

Patient

id, user_id, DOB, gender, blood_group, emergency_contact

Doctor

id, user_id, specialization, qualification, experience, registration_details

MedicalRecord

id, patient_id, type, title, date, file_url, extracted_text, created_at

AIAnalysis

id, record_id, summary, findings, abnormalities, created_at

Appointment

id, patient_id, doctor_id, date, time, type, status

Prescription

id, patient_id, doctor_id, appointment_id, medicines, instructions, created_at

Consent

id, patient_id, doctor_id, record_id, permission, granted_at, expires_at, status

AuditLog

id, user_id, action, resource, timestamp, metadata

28. Complete User Workflow

Patient Workflow

```
Register
   ↓
Login
   ↓
Create Health Profile
   ↓
Upload Previous Reports
   ↓
AI Processes Reports
```

```
   ↓
Medical Timeline Created
   ↓
Search Doctor
   ↓
Book Appointment
   ↓
Online / Offline Consultation
   ↓
Doctor Reviews Authorized Records
   ↓
Consultation
   ↓
Digital Prescription
   ↓
Record Added to Timeline
```

Doctor Workflow

```
Doctor Login
     ↓
Dashboard
     ↓
View Appointments
     ↓
Open Patient
     ↓
Check Consent
     ↓
View Medical Timeline
     ↓
Read AI Summary
     ↓
Open Original Reports
     ↓
Conduct Consultation
     ↓
Add Clinical Notes
     ↓
Generate Prescription
     ↓
Save Consultation
```

29. Innovation

- **EMR:** Centralized patient medical history.
- **AI:** Converts complex documents into structured summaries.
- **Timeline:** Creates longitudinal healthcare understanding.
- **Healthcare Access:** Connects records with appointment and consultation services.
- **Consent:** Supports patient-controlled sharing.
- **Interoperability:** Designed for future digital-health ecosystem integration.
- **Doctor-Centric Intelligence:** Reduces time required to understand previous medical information.

30. Unique Selling Proposition

Primary: “From scattered medical reports to one intelligent, connected health journey.”

Alternative: “Your complete medical history — organized, intelligent and available when you need it.”

31. Existing Problems vs Proposed Solution

- Scattered records → Centralized digital record
- Paper reports → Digital uploads

- Long documents → AI summaries
- Complex terminology → Patient-friendly explanations
- Doctor lacks previous context → Timeline + authorized history
- Long waiting times → Appointment scheduling
- Difficulty reaching specialists → Doctor discovery + telemedicine
- Repeated history-taking → Structured medical timeline
- Uncontrolled record sharing → Consent management
- Disconnected systems → Interoperability-ready architecture

32. Differentiation From SIH 2026 Problem Statement 26047

SIH 26047 — Patient Case-Taking Software primarily focuses on digitizing and improving the clinical history/case-taking process.

Our EMR concept is broader: longitudinal EMR, report storage, AI report analysis, medical timeline, doctor access, patient access, appointment booking, telemedicine, digital prescriptions, consent-based sharing and future interoperability.

Strong positioning: “We created an intelligent longitudinal EMR platform that converts fragmented medical documents and consultation records into a structured patient health timeline and connects that information with appointment, consultation and healthcare-access services.”

If the project is presented against SIH 26047, it should not be described as merely another case-taking application.

33. MVP for Hackathon

Must Have

- Patient login
- Doctor login
- Patient profile
- Medical report upload
- AI report summarization
- Medical timeline
- Doctor dashboard
- Authorized record access
- Appointment booking
- Digital consultation record
- Prescription generation
- Basic consent/access mechanism

Good to Have

- Telemedicine video integration
- OCR
- Voice-based history entry
- Multilingual support

- Notifications

Future

- ABDM integration
- ABHA integration
- FHIR interoperability
- Hospital integration
- Advanced clinical decision support

34. Hackathon Demo Flow

- Patient creates account.
- Patient uploads three previous reports.
- AI analyzes the reports.
- Dashboard creates the medical timeline.
- Patient searches for a doctor.
- Patient books an appointment.
- Doctor opens the patient profile.
- Doctor sees AI summary, timeline and original reports.
- Doctor conducts consultation.
- Doctor creates digital prescription.
- Consultation is automatically added to patient history.

Final Demo Statement: “The patient's entire healthcare journey has now become a connected digital record rather than a collection of disconnected documents.”

35. Future Scope

- Basic EMR
- AI report analysis
- Telemedicine
- Multilingual healthcare assistance
- ABDM interoperability
- FHIR-based healthcare data exchange
- Hospital/diagnostic-lab integration
- Advanced clinical decision-support tools
- Governed and de-identified population-level analytics

36. Multilingual & Accessibility

Future versions can support English, Hindi, Telugu, Tamil, Kannada, Malayalam, Bengali, Marathi and Odia. The platform can also support large text, voice input/output, multilingual interfaces and mobile-first low-bandwidth operation.

37. AI Safety

AI Can

- Summarize
- Organize
- Extract information
- Explain terminology
- Highlight information present in documents
- Create timelines
- Generate questions for doctor discussion

AI Should Not

- Independently diagnose diseases
- Replace doctors
- Prescribe medicines
- Change treatment without authorization
- Hide the original report
- Present uncertain conclusions as facts

38. Scalability

```
Patients
    ↓
Doctors
    ↓
Clinics
    ↓
Hospitals
    ↓
Diagnostic Labs
    ↓
Pharmacies
    ↓
Digital Health Ecosystem
```

39. Business Model — Future

- B2C premium patient features
- B2B subscriptions for clinics/hospitals
- B2B2C healthcare partnerships
- EMR SaaS
- Enterprise integrations

The platform should not rely on selling sensitive patient data.

40. Social Impact

- Healthcare accessibility
- Continuity of care
- Doctor productivity
- Patient awareness

- Digital-health adoption
- Rural healthcare access
- Telemedicine accessibility
- Medical record portability

41. Key Performance Indicators

- Reduction in time required to review patient history
- Report processing time
- Summary completeness
- Appointment booking time
- Number of patients/doctors/reports
- Unauthorized-access attempts detected
- Audit-log coverage

42. Project Success Criteria

Doctor: Open patient → understand previous history → review relevant reports → consult → create prescription → save consultation.

Patient: Store records → understand reports → find doctor → book consultation → receive prescription → maintain continuous health history.

43. Final Project Identity

- **Problem:** Fragmented medical data + long waiting times + limited healthcare accessibility
- **Solution:** AI-powered EMR + medical timeline + healthcare access platform
- **Users:** Patients + Doctors + Clinics + Hospitals
- **Core Technology:** React + Django REST + PostgreSQL + AI/NLP + OCR + Secure APIs
- **Core Innovation:** Transforming medical documents into an intelligent longitudinal health record
- **Security:** Authentication + authorization + consent + encryption + audit logs
- **Future:** FHIR + ABDM/ABHA interoperability + hospital/lab integration

Ultimate Goal: Build a connected digital healthcare ecosystem where the right medical information reaches the right healthcare professional at the right time — with the patient's knowledge and consent.

44. One-Minute Project Explanation

Our project is an AI-powered Electronic Medical Record platform designed to solve the problem of fragmented healthcare data. Today, a patient's medical information may be scattered across hospitals, clinics, laboratories, prescriptions and paper or digital reports. This makes it difficult for patients to maintain their complete history and for doctors to quickly understand previous medical conditions and treatments.

Our platform creates a longitudinal digital health record where patients can securely store their reports, prescriptions and consultation history. AI analyzes uploaded medical documents and converts them into structured summaries and a chronological medical timeline.

Patients can use the platform to find doctors, book appointments and access online or offline consultations. During consultation, doctors can access the patient's authorized medical history and reports, add clinical notes and generate digital prescriptions.

A consent-based access mechanism ensures that patients remain in control of their medical information. The architecture is also designed with future interoperability and India's digital-health ecosystem in mind.

In simple terms: We are transforming scattered medical documents into one intelligent, connected and patient-controlled healthcare journey.

45. Elevator Pitch

We are building a patient-centric AI-powered EMR platform that transforms fragmented medical records into an intelligent longitudinal health timeline and connects it with doctor discovery, appointment booking and telemedicine — while keeping patient consent and data security at the center.

46. Core Tagline

“One Patient. One Health Timeline. One Connected Healthcare Journey.”